PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

26. A circular disk of radius R meter and mass M kg is rotating around the axis perpendicular to the disk. An external torque is applied to the disk such that $\theta(t) = 5t^2 - 8t$, where $\theta(t)$ is the angular position of the rotating disc as a function of time t.

How much power is delivered by the applied torque, when $t = 2$ s?

(1) $72MR^2$ (2) $8MR^2$

(3) $108MR^2$ (4) $60MR^2$

Answer (4)

Sol. $I = \frac{MR^2}{2}$

$\omega = 10t - 8 = 12$ for $t = 2$

$\alpha = 10$

$P = \tau\omega = \frac{MR^2}{2} \times 10 \times 12$

$= 60MR^2$

27. A plane electromagnetic wave of frequency 20 MHz travels in free space along the $+x$ direction. At a particular point in space and time, the electric field vector of the wave is $E_y = 9.3\ \text{Vm}^{-1}$. Then, the magnetic field vector of the wave at that point is

(1) $B_Z = 1.55 \times 10^{-8}$ T

(2) $B_Z = 6.2 \times 10^{-8}$ T

(3) $B_Z = 9.3 \times 10^{-8}$ T

(4) $B_Z = 3.1 \times 10^{-8}$ T

Answer (4)

Sol. $E = CB$

$B = \frac{9.3}{3 \times 10^8}$

$= 3.1 \times 10^{-8}$

28. The width of one of the two slits in Young's double slit experiment is d while that of the other slit is xd. If the ratio of the maximum to the minimum intensity in the interference pattern on the screen is 9 : 4 then what is the value of x?

(Assume that the field strength varies according to the slit width.)

(1) 4 (2) 5

(3) 3 (4) 2

Answer (2)

Sol. $A \propto$ width of slit

$\Rightarrow A_1 = A, A_2 = xA$

$\frac{I_{max}}{I_{min}} = \frac{(A_1 + A_2)^2}{(A_1 - A_2)^2} = \frac{(x+1)^2}{(x-1)^2} = \frac{9}{4}$

$2x + 2 = 3x - 3$

$x = 5$

29. The energy of a system is given as $E(t) = \alpha^3 e^{-\beta t}$, where t is the time and $\beta = 0.3\ \text{s}^{-1}$. The errors in the measurement of α and t are 1.2% and 1.6%, respectively. At $t = 5$ s, maximum percentage error in the energy is :

(1) 6%

(2) 4%

(3) 8.4%

(4) 11.6%

Answer (1)



Sol. $dE = 3\alpha^2 d\alpha\, e^{-\beta t} + \alpha^3 e^{-\beta t}(-\beta)dt$

$$\frac{dE}{E} = \frac{3d\alpha}{\alpha} + (-\beta)dt$$

$$\frac{\Delta E}{E} = \frac{3\Delta\alpha}{\alpha} + \beta\Delta t$$

(%) $\Delta t = 1.6\% \times 5 = 8\%$

$$\%\left(\frac{3\Delta\alpha}{\alpha}\right) = 3 \times 1.2\% = 3.6\%$$

Putting values we get

$\Rightarrow$ $3.6 + 8 \times 0.3 = 6\%$

30. A massless spring gets elongated by amount x_1 under a tension of 5 N. Its elongation is x_2 under the tension of 7 N. For the elongation of $(5x_1 - 2x_2)$, the tension in the spring will be,

(1) 20 N (2) 11 N

(3) 39 N (4) 15 N

Answer (2)

Sol. As $T = kx$

$$x_1 = \frac{5}{k} \text{ \& } x_2 = \frac{7}{k}$$

Then $(5x_1 - 2x_2)k = 5 \times 5 - 2 \times 7 = 11$ N

31. Water flows in a horizontal pipe whose one end is closed with a valve. The reading of the pressure gauge attached to the pipe is P_1. The reading of the pressure gauge falls to P_2 when the valve is opened. The speed of water flowing in the pipe is proportional to

(1) $\sqrt{P_1 - P_2}$ (2) $(P_1 - P_2)^4$

(3) $P_1 - P_2$ (4) $(P_1 - P_2)^2$

Answer (1)

Sol. $\boxed{\quad P_1 \quad} P_1$

$\quad P_2 \quad \Rightarrow P_1$

Using Bernoulli's theorem

$$P_1 - P_2 = \frac{1}{2}\rho v^2$$

$$v \propto \sqrt{P_1 - P_2}$$

32. In photoelectric effect an em-wave is incident on a metal surface and electrons are ejected from the surface. If the work function of the metal is 2.14 eV and stopping potential is 2 V, what is the wavelength of the em-wave?

(Given hc = 1242 eVnm where h is the Planck's constant and c is the speed of light in vaccum.)

(1) 300 nm (2) 400 nm

(3) 600 nm (4) 200 nm

Answer (1)

Sol. $\phi = 2.14$

$V_S = 2$ V

Using photoelectric equation.

$$\frac{hc}{\lambda} = 2.14 + 2 = 4.14 \text{ eV}$$

$$\lambda = \frac{1242}{4.14} = 300 \text{ nm}$$

33. Water of mass m gram is slowly heated to increase the temperature from T_1 to T_2. The change in entropy of the water, given specific heat of water is 1 Jkg^{-1} K^{-1}, is

(1) $m(T_2 - T_1)$ (2) $m \ln\left(\frac{T_2}{T_1}\right)$

(3) $m \ln\left(\frac{T_1}{T_2}\right)$ (4) Zero

Answer (2)

Sol. $dQ = msdT$

$$ds = \frac{dQ}{T} = \frac{msdT}{T}$$

$$\Delta s = ms \ln\frac{T_2}{T_1}$$

$$\Delta s = ms \ln\frac{T_2}{T_1} \text{ as } s = 1$$



34. A ball having kinetic energy KE, is projected at an angle of 60° from the horizontal. What will be the kinetic energy of ball at the highest point of its flight?

(1) $\frac{(KE)}{2}$ (2) $\frac{(KE)}{4}$

(3) $\frac{(KE)}{16}$ (4) $\frac{(KE)}{8}$

Answer (2)

Sol. Let speed at projection is v_0

at highest point it would be $v_0\cos\theta$

as $KE \propto v^2$

$KE_{Highest} = KE_{Projection}\cos^2\theta$

$= \frac{1}{4} KE_{Projection}$

35. Match **List - I** with **List - II**.

	List - I		**List - II**
(A)	Permeability of free space	(I)	$[M L^2T^{-2}]$
(B)	Magnetic field	(II)	$[MT^{-2}A^{-1}]$
(C)	Magnetic moment	(III)	$[MLT^{-2}A^{-2}]$
(D)	Torsional constant	(IV)	$[L^2A]$

Choose the **correct** answer from the options given below

(1) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

(2) (A)-(II), (B)-(I), (C)-(III), (D)-(IV)

(3) (A)-(I), (B)-(IV), (C)-(II), (D)-(III)

(4) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

Answer (1)

Sol. (A) $B = \frac{\mu_0 i}{2\pi r}$ & $qvB = F$

$$\mu_0 \equiv \frac{Br}{i} \equiv \frac{Fr}{qvi} \equiv \frac{MLT^{-2} \times L}{A^2TLT^{-1}} \equiv MLT^{-2}A^{-2} \quad \text{(III)}$$

(B) $B \equiv \frac{F}{qv} \equiv \frac{MLT^{-2}}{ATLT^{-1}} = MT^{-2}A^{-1}$ (II)

(C) $M = iA \equiv AL^2$ (IV)

(D) $\tau = C\theta \Rightarrow C \equiv ML^2T^{-2}$ (I)

36. A galvanometer having a coils resistance 30 Ω need 20 mA of current for full-scale deflection. If a maximum current of 3 A is to be measured using this galvanometer, the resistance of the shunt to be added to the galvanometer should be $\frac{30}{X}\Omega$, where X is

(1) 596 (2) 149

(3) 298 (4) 447

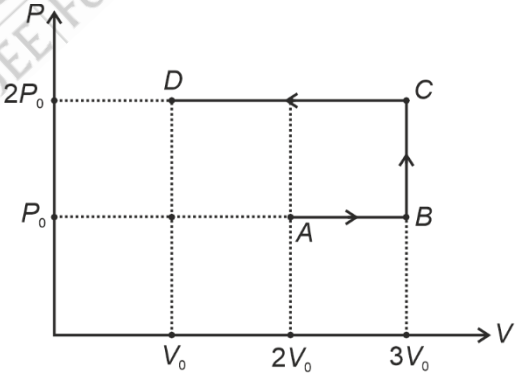
Answer (2)

Sol. $(I - I_g)\,R = I_g G$

$(3 - 0.02) \times R = 0.02 \times G \Rightarrow R = \frac{30}{149}$

$\Rightarrow$ 149 = Required X

37



Using the given P-V diagram, the work done by an ideal gas along the path $ABCD$ is

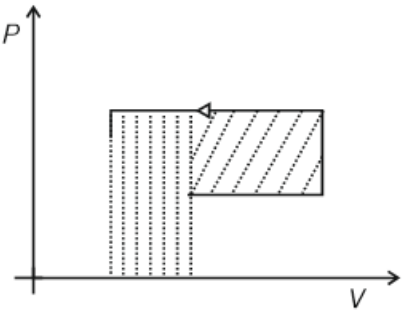
(1) $-3\,P_0V_0$ (2) $-4\,P_0V_0$

(3) $4\,P_0V_0$ (4) $3\,P_0V_0$

Answer (1)



Sol.

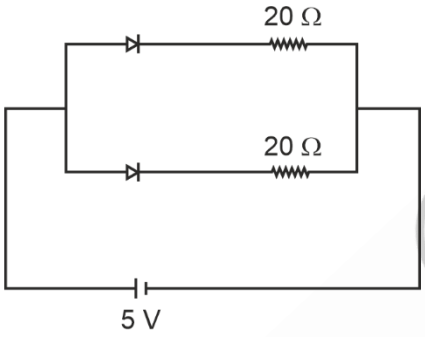


Area under graph will be magnitude of graph and being counterclockwise graph it would be negative

Area = $2P_0 \times V_0 + P_0V_0 = 3P_0V_0$

$W = -3P_0V_0$

38. What is the current through the battery in the circuit shown below?



(1) 1.5 A (2) 1.0 A

(3) 0.5 A (4) 0.25 A

Answer (3)

Sol. Diodes are in forward bias therefore

$$i = \frac{V}{R_{eq}} = \frac{5}{10} = 0.5 \text{ A}$$

39. If a satellite orbiting the Earth is 9 times closer to the Earth than the Moon, what is the time period of rotation of the satellite? Given rotational time period of Moon = 27 days and gravitational attraction between the satellite and the moon is neglected.

(1) 1 day (2) 27 days

(3) 3 days (4) 81 days

Answer (1)

Sol. $T^2 \propto r^3$ from Kepler's 3rd law

$$\frac{27^2}{T^2} = \frac{r^3}{(r/g)^3}$$

$$\frac{27^2}{9^3} = T^2$$

$$\frac{27 \times 27}{9 \times 9 \times 9} = T^2 \Rightarrow T = 1 \text{ day}$$

40. Two point charges –4 μC and 4 μC, constituting an electric dipole, are placed at (–9, 0, 0) cm and (9, 0, 0) cm in a uniform electric field of strength 10^4 NC^{-1}. The work done on the dipole in rotating it from the equilibrium through 180° is

(1) 14.4 mJ (2) 18.4 mJ

(3) 16.4 mJ (4) 12.4 mJ

Answer (1)

Sol. $U = -\vec{P} \cdot \vec{E}$

Therefore $W = 2PE$

$= 2 \times 4 \times 10^{-6} \times 18 \times 10^{-2} \times 10^4$

$= 144 \times 10^{-4}$

$= 14.4$ mJ

41. Given below are two statements. One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Reason (A): The binding energy per nucleon is found to be practically independent of the atomic number A, for nuclei with mass numbers between 30 and 170.

Reason (R): Nuclear force is long range.

In the light of the above statements, choose the correct answer from the options given below:

(1) (A) is true but (R) is false

(2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

(3) Both (A) and (R) are true and (R) is the correct explanation of (A)

(4) (A) is false but (R) is true

Answer (1)



Sol. From graph between B.E/N & A we can see BE/N is almost constant ⇒ correct

Reason ⇒ incorrect as nuclear forces are short range forces.

42. Two charges 7 μC and –4 μC are placed at (–7 cm, 0, 0) and (7 cm, 0, 0) respectively. Given, $\varepsilon_0 = 8.85 \times 10^{-12}\ C^2\ N^{-1}\ m^{-2}$, the electrostatic potential energy of the charge configuration is:

(1) – 1.5 J (2) – 1.2 J

(3) – 2.0 J (4) – 1.8 J

Answer (4)

Sol. $U = \frac{kq_1q_2}{r}$

$= -\frac{9\times10^9\times7\times4\times10^{-12}}{14\times10^{-2}}$

$= -18 \times 10^{-1}$

$= -1.8$ J

43. A concave mirror of focal length f in air is dipped in a liquid of refractive index μ. Its focal length in the liquid will be :

(1) $\frac{f}{(\mu-1)}$ (2) $\frac{f}{\mu}$

(3) f (4) μf

Answer (3)

Sol. Focal length of mirror is independent of refractive index.

44. The equation of a transverse wave travelling along a string is

$y(x, t) = 4.0 \sin[20 \times 10^{-3}\ x + 600t]$ mm, where x is in mm and t is in second. The velocity of the wave is :

(1) – 60 m/s (2) + 30 m/s

(3) – 30 m/s (4) + 60 m/s

Answer (3)

Sol. $k = 20 \times 10^{-3}\ mm^{-1} = 20\ m^{-1}$

$w = 600\ s^{-1}$

$v = \frac{w}{k} = \frac{600}{20} = 30$ m/s

and x & t carry same sign

Therefore v = –30 m/s

45. The refractive index of the material of a glass prism is $\sqrt{3}$. The angle of minimum deviation is equal to the angle of the prism. What is the angle of the prism?

(1) 48° (2) 50°

(3) 60° (4) 58°

Answer (3)

Sol. $\mu = \frac{\sin(A+\delta m/2)}{\sin A/2}$ and $\delta_m = A$

$\sqrt{3} = \frac{\sin A}{\sin A/2} = 2\cos A/2$

$\cos A/2 = \cos 30°$

$A = 60°$

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

46. A time varying potential difference is applied between the plates of a parallel plate capacitor of capacitance 2.5 μF. The dielectric constant of the medium between the capacitor plates is 1. It produces an instantaneous displacement current of 0.25 mA in the intervening space between the capacitor plates, the magnitude of the rate of change of the potential difference will be _____ Vs^{-1}.

Answer (100)

Sol. $q = CV$

Differentiating

$i = \frac{CdV}{dt}$

$\Rightarrow \frac{dV}{dt} = \frac{0.25\times10^{-3}}{2.5\times10^{-6}} = \frac{1000}{10} = 100$



47. An air bubble of radius 1.0 mm is observed at a depth of 20 cm below the free surface of a liquid having surface tension 0.095 J/m^2 and density 10^3 kg/m^3. The difference between pressure inside the bubble and atmospheric pressure is _____ N/m^2.

(Take g = 10 m/s^2)

Answer (2190)

Sol. $\Delta p = h\rho g + \frac{2s}{r}$

$= 0.2\times10^3\times10 + \frac{2\times0.095}{10^{-3}}$

= 2000 + 190 = 2190 Pa

48. In a series LCR circuit, a resistor of 300 Ω, a capacitor of 25 μF and an inductor of 100 mH are used. For maximum current in the circuit, the angular frequency of the ac source is _____ × 10^4 radians s^{-1}.

Answer (2)

Sol. $\omega = \frac{1}{\sqrt{LC}}$ is resonance condition

$= \frac{1}{\sqrt{100\times10^{-3}\times25\times10^{-9}}}$

$= \frac{10^6}{5\times10}$

= 2 × 10^4

49. A satellite of mass $\frac{M}{2}$ is revolving around earth in a circular orbit at a height of $\frac{R}{3}$ from earth surface. The angular momentum of the satellite is $M\sqrt{\frac{GMR}{x}}$. The value of x is _____, where M and R are the mass and radius of earth, respectively. (G is the gravitational constant)

Answer (3)

Sol. $L = mvr$

$L = m\sqrt{\frac{GM}{R}}R$

$= m\sqrt{GMR}$

$= \frac{M}{2}\sqrt{GM\frac{4R}{3}}$

$= M\sqrt{\frac{GMR}{3}}$

x = 3 (But for comparable mass, this solution is not applicable)

Alternate solution:

$m_1r_1 = m_2r_2 \Rightarrow r_1 = \frac{8R}{9}$

$\frac{GM\frac{M}{2}}{\left(\frac{4R}{3}\right)^2} = \frac{M}{2}\omega^2\frac{8R}{9}$

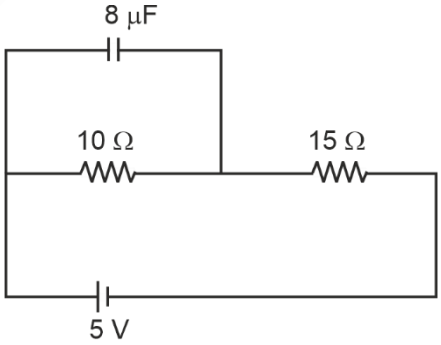
$\omega = \sqrt{\frac{81GM}{128R^3}}$

$L = I\omega = \frac{M}{2}\left(\frac{8R}{9}\right)^2\omega$

$= M\sqrt{GMR\frac{8}{81}}$

$\Rightarrow x \approx 10$

50. At steady state the charge on the capacitor, as shown in the circuit below, is _____ μC.



Answer (16)

Sol. $V_{10\,\Omega} = V_C = \frac{5}{25}\times10 = 2$ V

Q = 8 × 2 = 16 μC



CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

51. Consider a binary solution of two volatile liquid components 1 and 2. x_1 and y_1 are the mole fractions of component 1 in liquid and vapour phase, respectively. The slope and intercept of the linear plot of $\frac{1}{x_1}$ vs $\frac{1}{y_1}$ are given respectively as:

(1) $\frac{P_1^0}{P_2^0}, \frac{P_1^0 - P_2^0}{P_2^0}$ (2) $\frac{P_1^0}{P_2^0}, \frac{P_2^0 - P_1^0}{P_2^0}$

(3) $\frac{P_2^0}{P_1^0}, \frac{P_2^0 - P_1^0}{P_2^0}$ (4) $\frac{P_2^0}{P_1^0}, \frac{P_1^0 - P_2^0}{P_2^0}$

Answer (2)

Sol. Given mole fraction of liquid 1 in liquid and vapour phase is x_1 and y_1 respectively.

$P_1 = P_T y_1$ (DLPP)

$P_1 = P_1^0 x_1$ (Raoult's law)

$P_1 = P_T y_1 = P_1^0 x_1$

$$\frac{P_T}{x_1} = \frac{P_1^0}{y_1}$$

$$\frac{P_2^0 + x_1(P_1^0 + P_2^0)}{x_1} = \frac{P_1^0}{y_1}$$

$$\frac{P_2^0}{x_1} + (P_1^0 + P_2^0) = \frac{P_1^0}{y_1}$$

$$\frac{P_2^0}{x_1} = \frac{P_1^0}{y_1} - (P_1^0 + P_2^0)$$

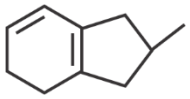
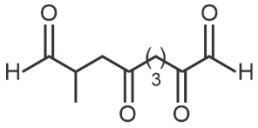
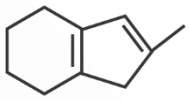
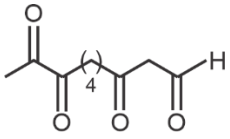
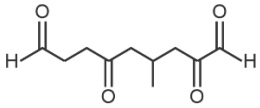
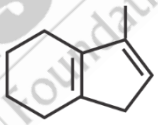
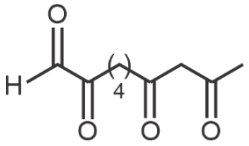
$$\frac{1}{x_1} = \frac{1}{y_2}\left(\frac{P_1^0}{P_2^0}\right) - \frac{(P_1^0 + P_2^0)}{P_2^0}$$

$$\frac{1}{x_1} = \frac{1}{y_2}\left(\frac{P_1^0}{P_2^0}\right) + \frac{(P_2^0 - P_1^0)}{P_2^0}$$

$y = mx + c$

Slope = $\frac{P_1^0}{P_2^0}$; Intercept = $\frac{P_2^0 - P_1^0}{P_2^0}$

52. Match List-I with List-II

List-I (Isomers of $C_{10}H_{14}$)	List-II (Ozonolysis product)
(A) 	(I) 
(B) 	(II) 
(C) 	(III) 
(D) 	(IV) 

Choose the correct answer from the options given below:

(1) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

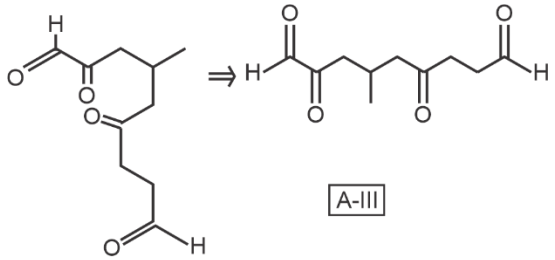
(2) (A)-(I), (B)-(IV), (C)-(III), (D)-(II)

(3) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(4) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

Answer (1)

Sol.

(A) 

A-III

